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PD-L1 Scoring Models for Non-Small Cell Lung Cancer in China: Current Status, AI-Assisted Solutions and Future Perspectives

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ABSTRACT

Immunotherapy has revolutionized the diagnosis and treatment model for patients with advanced non-small cell lung cancer (NSCLC). Numerous clinical trials and real-world reports have confirmed that PD-L1 status is a key factor for the successful use of immunotherapy in NSCLC, by predicting clinical outcomes and identifying patients most likely to benefit from this treatment. Therefore, accurate and standardized evaluation of PD-L1 expression is crucial. Currently, PD-L1 testing in China faces several challenges, including a heavy pathologist workload, a shortage of highly trained pathologists plus the inadequate capacity of diagnostic laboratories, confusion around different scoring methods, cut-off values, and indications, and limited concordance between PD-L1 assays. In this review, we summarize the current status and limitations of PD-L1 testing for patients with NSCLC in China and discuss recent progress in artificial intelligence-assisted PD-L1 scoring. Our review aims to support improvements in clinical PD-L1 testing practice and optimization of the prognosis and outcomes of immunotherapy in this patient population.

1 | Introduction

Lung cancer causes significant morbidity and mortality, in China and worldwide [1]. In China, lung cancer is the leading cause of cancer-related mortality, resulting in around one-fourth of all cancer-related deaths, and it remains a major public health problem [2]. Non-small cell lung cancer (NSCLC) is the predominant type of lung cancer, accounting for 80%–85% of all lung cancer cases globally [3]. In recent years, immunotherapy with immune checkpoint inhibitors

(ICIs) has revolutionized the diagnosis and treatment model for patients with advanced NSCLC. By binding to the programmed cell death protein 1 (PD-1) receptor on the surface of immune cells such as T cells and B cells, programmed cell death-ligand 1 (PD-L1) (expressed on tumor cells) helps tumors evade immune surveillance and escape destruction by the immune system [4, 5]. Numerous clinical trials and real-world reports have confirmed that ICIs have promising antitumor activity and acceptable toxicity in the treatment of patients with advanced or metastatic NSCLC, and PD-L1

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status was confirmed as the ideal biomarker to predict clinical outcomes and identify patients likely to respond to ICI therapy [6]. While not all patients with NSCLC respond to treatment with ICIs, patients with tumors that express high levels of PD-L1 are more likely to respond to ICIs and have better clinical outcomes [7].

Accurate, reproducible, and standardized evaluation of PD-L1 expression in tumor tissues is crucial for the reliable use of PD-L1 as a biomarker to predict treatment outcomes and to select patients for immunotherapy. Currently, PD-L1 testing in China faces several challenges, including heavy pathologist workload, a shortage of highly trained pathologists, inadequate capacity of diagnostic laboratories in certain provinces, confusion around different scoring methods, cut-off values and indications, and limited concordance between PD-L1 assays. In this article, we review the current status and limitations of standard PD-L1 testing for patients with NSCLC in China and discuss recent progress in artificial intelligence-assisted PD-L1 scoring. Our review aims to support improvements in clinical PD-L1 testing practice and optimization of the prognosis and outcomes of immunotherapy in this patient population.

2 | Role and Challenges of Immunotherapy and PD-L1 Testing in NSCLC in China

2.1 | Essential Role and Challenges of Immunotherapy in NSCLC in China

Immunotherapy has become an essential part of NSCLC treatment in China [8]. Since nivolumab was first approved by the Chinese National Medical Products Administration (NMPA) for the treatment of NSCLC in 2018 and followed by pembrolizumab in 2019, several other anti-PD-1/PD-L1 therapies have been approved by the NMPA for the first- or second-line treatment of NSCLC. To date, the NMPA has approved 11 anti-PD-1/PD-L1 monoclonal antibodies (mAbs) for NSCLC, including eight anti-PD-1 antibodies (camrelizumab, nivolumab, pembrolizumab, sintilimab, tislelizumab, toripalimab, penpulimab [AK105], and serplulimab) [8–15] and three anti-PD-L1 antibodies (atezolizumab, durvalumab, and sugemalimab) [8, 9, 11, 16–18] (Table 1). Only atezolizumab and pembrolizumab have been approved for use as monotherapy in the first-line setting in selected patients (Table 1) [9].

Moreover, guidelines from the Chinese Society of Clinical Oncology (CSCO) also recommend immunotherapy for patients with NSCLC (Table 2). However, two recent studies investigating trends in the use of PD-1/PD-L1 inhibitors by oncologists in China showed that platinum-based chemotherapy is the preferred treatment for advanced NSCLC in the first-line setting [19] and less than 60% of Chinese clinicians prescribe PD-1/PD-L1 inhibitors in this setting [20]. This is partly because the objective response rates with ICI monotherapy are typically below 30% for previously treated patients [21], although superior efficacy of ICIs over chemotherapy or placebo has been reported [21–25], and combining ICIs with chemotherapy is associated with objective response rates of up to 50% [24]. In addition, clinicians tend to prefer the use of pembrolizumab monotherapy or

in combination with platinum doublet chemotherapy in patients with PD-L1 expression in $\geq 50\%$ of tumor cells [10], and patients with squamous histology are more likely to be prescribed PD-1/PD-L1 inhibitors than those with non-squamous histology in this setting [19]. The high cost of mAbs may also contribute to the lower-than-expected frequency of ICI use in Chinese patients with advanced NSCLC [10]. Therefore, efforts are needed to reinforce an accurate understanding of the rationale and clinical utility of ICIs in NSCLC among Chinese oncologists.

2.2 | Pivotal Role of PD-L1 Expression in Patient Selection

PD-L1 expression is a biomarker for tumor response and survival following treatment with anti-PD-1/PD-L1-based therapies, as shown in the KEYNOTE-407 [26] and KEYNOTE-189 trials [27]. Real-world evidence from China also supports an association between PD-L1 expression and survival following treatment with an ICI and chemotherapy [28, 29]. An artificial intelligence-based model that uses PD-L1 TPS to predict response to ICI treatment was clinically validated in patients with advanced NSCLC [30]. However, methodological variability, as well as the dynamic and heterogeneous nature of PD-L1 expression, have led to uncertainty in test selection and implementation in practice.

2.3 | Challenges in the Implementation of Clinical PD-L1 Testing in NSCLC in China

Following the approval of PD-L1/PD-1 inhibitors by the Chinese NMPA, five companion or complementary diagnostics have been approved (Table 3), all of which are based on the analysis of formalin-fixed paraffin-embedded (FFPE) tumor tissues using immunohistochemistry (IHC) [8]. The *Chinese Expert Consensus on Standards of PD-L1 Immunohistochemistry Testing for Non-small Cell Lung Cancer* published by the Chinese Anti-cancer Association provides recommendations for the selection of PD-L1 detection reagents and detection platforms [37]. While the 2023 CSCO guidelines recommend PD-L1 expression testing and *EGFR/ALK* mutation testing at the initial diagnosis of patients with advanced NSCLC [9], the rate of PD-L1 testing in China is low and varies from hospital to hospital [38]. In addition, the choice of PD-L1 test depends on the therapeutic agent being considered and the stage of NSCLC and can impact the interpretation of PD-L1 expression and subsequent treatment decisions [8]. Additionally, in many regions of China (e.g., Shanghai), PD-L1 testing, such as the 22C3 kit, is not covered by the national medical insurance and this greatly affects the utilization of PD-L1 testing. Regional differences in the utilization of the different PD-L1 kits have been observed between Asia and North America, with higher use of the 28–8 (29% vs. 12%), SP142 (38% vs. 20%), and SP263 (64% vs. 35%) clones in Asia (Figure 1) [39].

According to economic analyses, from the perspective of the Chinese healthcare system, PD-L1 testing is cost-effective in guiding the use of ICIs in patients with advanced NSCLC in the first-line but not in the second-line setting [40, 41]. From a practical perspective, 30%–40% of patients with advanced NSCLC in

TABLE 1 | Immune checkpoint inhibitors approved in China for the treatment of non-small cell lung cancer.

Name	Antibody class	Target	Indications
Atezolizumab [9]	Humanized IgG1 mAb	PD-L1	– As monotherapy for the first-line treatment of patients with metastatic NSCLC whose tumors have high PD-L1 expression (PD-L1 expression in $\geq 50\%$ of tumor cells or PD-L1 expression in tumor-infiltrating immune cells covering $\geq 10\%$ of the tumor area), without <i>EGFR</i> or <i>ALK</i> genomic tumor aberrations
Camrelizumab [9]	Humanized IgG4 mAb	PD-1	– In combination with pemetrexed and carboplatin as first-line treatment for patients with unresectable, locally advanced, or metastatic nonsquamous NSCLC, without <i>EGFR</i> or <i>ALK</i> genomic tumor aberrations
Durvalumab [11, 16, 17]	Humanized IgG1 mAb	PD-L1	– As consolidation treatment for patients with unresectable, stage III NSCLC whose disease has not progressed following concurrent platinum-based chemotherapy and radiation therapy
Nivolumab [10, 11]	Humanized IgG4 mAb	PD-1	– As monotherapy for the second-line treatment of patients with advanced nonsquamous or squamous NSCLC regardless of PD-L1 expression or <i>EGFR/ALK</i> status (if not received before) – As monotherapy for the third-line treatment of patients with advanced nonsquamous or squamous NSCLC regardless of PD-L1 expression or <i>EGFR/ALK</i> status (if not received before)
Pembrolizumab [9, 10]	Humanized IgG4 mAb	PD-1	– In combination with pemetrexed and platinum chemotherapy as a first-line treatment for patients with metastatic nonsquamous NSCLC, without <i>EGFR</i> or <i>ALK</i> genomic tumor aberrations – As monotherapy for the first-line treatment of patients with locally advanced or metastatic nonsquamous or squamous NSCLC expressing PD-L1 (TPS $\geq 1\%$), without <i>EGFR</i> or <i>ALK</i> genomic tumor aberrations – In combination with carboplatin and paclitaxel as first-line treatment for patients with metastatic squamous NSCLC
Sintilimab [9, 12]	Humanized mAb	PD-1	– In combination with pemetrexed/platinum as first-line treatment for patients with unresectable, locally advanced, or metastatic nonsquamous NSCLC – In combination with platinum/gemcitabine as first-line treatment for patients with unresectable, locally advanced, or metastatic squamous NSCLC
Sugemalimab [18]	Humanized IgG4 mAb	PD-L1	– In combination with pemetrexed and carboplatin as first-line treatment for patients with metastatic nonsquamous NSCLC without <i>EGFR</i> or <i>ALK</i> tumor mutations – In combination with paclitaxel and carboplatin as first-line treatment for patients with metastatic squamous NSCLC without <i>EGFR</i> or <i>ALK</i> tumor mutations
Tislelizumab [9]	Humanized IgG4 mAb	PD-1	– In combination with pemetrexed/platinum as first-line treatment for patients with nonsquamous NSCLC – In combination with carboplatin/paclitaxel or nab-paclitaxel as first-line treatment for patients with squamous NSCLC
Toripalimab [13]	Humanized IgG4 mAb	PD-1	– In combination with plus pemetrexed and platinum as first-line treatment for patients with advanced non-squamous NSCLC who do not harbor <i>EGFR</i> mutations or <i>ALK</i> fusions
Penpulimab [AK105] [14]	Fc-engineered IgG1 anti-PD-1 antibody	PD-1	– In combination with paclitaxel and carboplatin for the first-line treatment of adult patients with metastatic non-squamous, NSCLC
Serplulimab [15]	Humanized IgG4 mAb	PD-1	– In combination with carboplatin and albumin-bound paclitaxel for the first-line treatment of patients with unresectable locally advanced or metastatic squamous non-small cell lung cancer

Abbreviations: mAb, monoclonal antibody; NSCLC, non-small cell lung cancer; TPS, tumor proportion score.

TABLE 2 | Chinese Society of Clinical Oncology recommendations for the use of immunotherapy in non-small cell lung cancer.

Stage	Status/ characteristics	Recommended use	Immunotherapy regimen
Resectable Stage IIIA-B	Following successful resection	Adjuvant therapy	• Atezolizumab monotherapy (PD-L1 TC $\geq 1\%$) (Level 1 recommendation) • Pembrolizumab monotherapy (Level 2 recommendation)
		Neoadjuvant therapy	• Nivolumab combined with platinum-based therapy (Level 1 recommendation)
	N ₂ disease and unable to undergo radical resection	Consolidation therapy following chemoradiotherapy	• Durvalumab or sugemalimab (Level 1 recommendation)
Unresectable Stage IIIA-C	ECOG PS 0–1	Consolidation therapy following chemoradiotherapy	• Durvalumab or sugemalimab (Level 1 recommendation)
	ECOG PS 2	Same as recommendations for Stage IV without driver mutations (Level 2 recommendation)	
Stage IV non- squamous NSCLC without driver mutations	First-line, ECOG PS 0–1	Systemic treatment	• Atezolizumab monotherapy (PD-L1 TC score $\geq 50\%$ or IC $\geq 10\%$) (Level 1 recommendation) • Pembrolizumab (PD-L1 TPS $\geq 50\%$ or TPS 1%–49% [2A]) (Level 1 recommendation) • Pemetrexed plus platinum chemotherapy combined with pembrolizumab, camrelizumab, sintilimab, tislelizumab, atezolizumab, sugemalimab, or toripalimab (Level 1 recommendation) • Paclitaxel plus carboplatin + bevacizumab + atezolizumab (Level 2 recommendation) • Albumin bound paclitaxel plus carboplatin + bevacizumab + atezolizumab (Level 2 recommendation) • Two cycles of pemetrexed plus platinum-based chemotherapy + nivolumab + ipilimumab (Level 3 recommendation)
		Systemic treatment	• Nivolumab or tislelizumab, if not used in first line (Level 1 recommendation) • Pembrolizumab (PD-L1 TPS $\geq 1\%$) or atezolizumab (Level 2 recommendation)
		Systemic treatment	• Nivolumab (if not already used) (Level 1 recommendation)
	Second-line, ECOG PS 0–2	Systemic treatment	
	Third-line, ECOG PS 0–2	Systemic treatment	

Abbreviations: ECOG PS, Eastern Co-operative Oncology Group Performance Status; IC, immune cell; NSCLC: non-small cell lung cancer; PD-L1: programmed death ligand 1; TC: tumor cell; TPS: tumor proportion score.

China only have cytology specimens available for analysis [42], suggesting a potential limitation in obtaining adequate tissue samples for PD-L1 testing. A study also found that long-term storage of paraffin-embedded tissues leads to antigen loss, but this can be delayed by refrigerated storage at -80°C [43]. There are also differences in the genomic profiles of Chinese patients with NSCLC versus their Western counterparts, with a higher prevalence of EGFR mutations and ALK rearrangements and a lower frequency of KRAS mutations [44].

3 | Complex PD-L1 Testing Requires Experienced Pathologists

3.1 | Complexity of Domestic PD-L1 Testing

Companion diagnostics are mandatory tests that must be performed before treatment is initiated, and complementary tests aid the therapeutic decision process but are not required when prescribing the corresponding ICI [45]. As shown in Table 3, the

TABLE 3 | NMPA-approved assays for PD-L1 testing in non-small cell lung cancer in China.

Clone	PD-L1 assays [7–9, 31–36]				
	22C3 ^a	28–8 ^a	SP142 ^a	SP263 ^a	E1L3N
Platform	AutoStainer Link 48		Ventana Benchmark Ultra		Laboratory-developed platform
Scoring system	TPS	TPS	TC, IC	TC	TPS
Drug	Pembrolizumab Camrelizumab Sintilimab	Nivolumab	Atezolizumab	Durvalumab	Pembrolizumab
Companion diagnostic	✓	✓	✓	✓	✓
Complementary diagnostic			✓		

Abbreviations: IC, immune cell; NMPA, National Medical Products Administration; PD-L1, programmed death-ligand 1; TC, tumor cell; TPS, tumor proportion score.

^aApproved by both the US FDA and China NMPA.

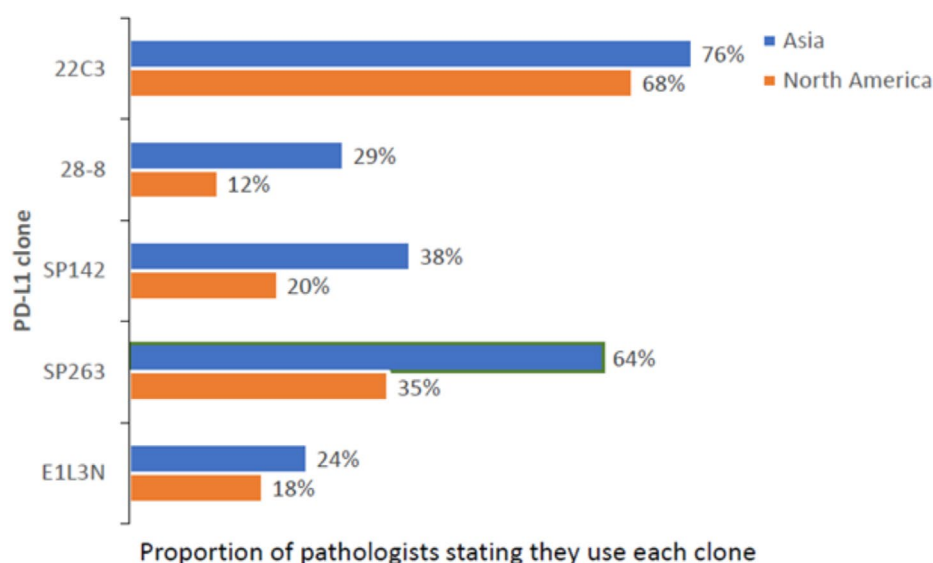


FIGURE 1 | Use of PD-L1 clones in Asia and North America based on a global survey of pathologists. The figure shows the self-reported use of different PD-L1 clones for PD-L1 testing among pathologists in Asia and North America. PD-L1, programmed death ligand 1. Drawn from data published by Mino-Kenudson et al. [39].

currently available PD-L1 assays differ in their antibody clones, platforms, scoring algorithms, and cutoff values for defining PD-L1 positivity. Therefore, pathologists must familiarize themselves with the specifications of each PD-L1 assay to correctly interpret PD-L1 testing results.

In terms of scoring algorithms, various metrics are used to estimate PD-L1 expression in NSCLC. Tumor proportion score (TPS), the scoring metric used for the 22C3 and 28–8 assays (used for pembrolizumab, camrelizumab, sintilimab, and nivolumab) represents the percentage of viable tumor cells showing partial or complete membrane staining for PD-L1 at any intensity [31, 46]. For the 22C3 assay, a TPS score of $\geq 1\%$ indicates PD-L1 expression, and a TPS score of $\geq 50\%$ indicates high PD-L1 expression [32, 46–49]. In contrast, for the 28–8 assay, TPS $\geq 25\%$ is used to determine PD-L1 positivity. Tumor cell (TC) score represents the percentage of tumor cells with positive staining

in the total tumor area and is the scoring method used for the SP263 and SP142 assays (used for atezolizumab and durvalumab) [31, 46]. PD-L1 positivity using TC score is defined as TC $\geq 50\%$ [46]. The immunocyte (IC) score is used as a scoring method for the SP142 assay to report the percentage of immune cells with positive staining in the total tumor area [31, 46]. There are no widely accepted cutoff values for determining PD-L1 expression on immune cells. However, NSCLC samples with IC $\geq 10\%$ are typically considered PD-L1-positive [46].

3.2 | High Demand for Experienced Pathologists

PD-L1 scoring accuracy and repeatability depend on expert experience. Manual assessment of IHC is the current gold standard, and inter-observer variability is inevitable [50]. However, expert pathologists have shown higher concordance in PD-L1

scoring in NSCLC than less experienced pathologists [51], suggesting that specialized training is needed to maintain consistency and quality of interpretation between pathologists [46]. Indeed, studies have shown that obtaining formal training on TPS can increase concordance in PD-L1 testing in NSCLC [52, 53]. Nonetheless, highly experienced pathologists are scarce, and providing specialized training to pathologists is not always feasible. While implementing continuing education and training could improve pathologists' ability to interpret PD-L1 scores, the complexity of the Chinese healthcare system (e.g., from primary hospitals to tertiary hospitals) [54], the high costs of training [55], and the heavy workload of Chinese pathologists [56, 57] make it challenging to provide training to all pathologists.

4 | Clinical PD-L1 Assay Limitations

4.1 | Limitations in the Implementation of Clinical PD-L1 Testing

PD-L1 assays are routinely used in clinical practice to determine PD-L1 expression in patients with NSCLC. However, differences in assay antibodies, platforms, scoring algorithms, and cutoff values can lead to variability in PD-L1 expression results and may influence patient eligibility for immunotherapy [58–60]. Expert panel members of the College of American Pathologists (CAP) concluded that the balance of effects (benefits and harms) did not favor either companion diagnostic (CDx) assays or laboratory-developed tests (LDTs) [61]. However, to date, the available PD-L1 IHC assays have not been clinically validated for cross-utilization [58].

There is also difficulty in determining what constitutes an optimal specimen for testing in patients with advanced NSCLC, in terms of testing of the primary tumor and/or metastatic sites, using small biopsy samples (core biopsies, cytology specimens, and endobronchial biopsies) and/or resection specimens. TPS scores obtained from the primary tumor and synchronous metastatic lesions are discordant versus resected specimens approximately 20% to 30% of the time, at both the 1% and 50% cutoffs [62–65]. Furthermore, intratumoral heterogeneity of PD-L1 expression is common in resection specimens, leading to different results obtained from small biopsy samples [27, 66, 67].

Finally, a lack of uniform guidelines for scoring PD-L1 expression across different laboratories and institutions also makes it difficult to compare PD-L1 test results and may lead to inconsistent treatment decisions. Therefore, efforts are needed to harmonize testing methodologies in China and internationally.

4.2 | Manual Inspection Is Time Consuming and Labor Intensive

Standard PD-L1 scoring methods require pathologists to manually examine stained tissue samples under a microscope, which is time-consuming and labor intensive [68]. Because of the large population (1.37 billion) of China and the increasing disease incidence and detection rate for lung cancer, the workload of pathologists in China has become extremely heavy, and the requirements for test quality, turnaround time, and reporting cycle are increasingly high [57].

4.3 | Multiple Metrics to Improve Predictive Performance

Current PD-L1 assays rely solely on PD-L1 expression as a predictive biomarker and may overlook other inhibitory receptors and ligands involved in immune regulation [7–9, 31–35]. Combining PD-L1 expression with additional biomarkers, such as tumor mutational burden (TMB), CD8+ T cell infiltration, and HLA class I expression, may provide a more comprehensive understanding of the immunological status of a tumor. However, integrating multiple biomarkers further increases the time required for manual scoring of IHC-stained tissue slides.

5 | Strategies for Optimizing Testing Practices

Several efforts have been made to overcome the limitations of standard PD-L1 scoring methods. Guidelines and consensus statements may help to standardize the assay process, including antibody/platform selection, scoring methods, and cutoff values, to ensure consistent and accurate quantification of PD-L1 and other biomarkers [69, 70]. In this regard, the recent publication of a Chinese expert consensus on PD-L1 testing for NSCLC [37, 71] or solid tumors [72] provides an initial step toward standardized testing and reporting. Additionally, the implementation of training programs, guidelines, and quality control initiatives can help minimize variability and ensure consistent interpretation of PD-L1 testing results. Several international projects have also been launched to standardize IHC-based PD-L1 assays. The most prominent is the Blueprint PD-L1 Immunohistochemistry Comparability Project, led by the International Association for the Study of Lung Cancer [73, 74].

Moreover, sampling multiple tumor regions and longitudinal retesting could help overcome the challenge of heterogeneity in PD-L1 expression in NSCLC [75]. Nonetheless, this may be technically challenging and time-consuming, further increasing the workload of Chinese pathologists. In addition, extensive pathologist training and the use of standard protocols and thresholds have been shown to increase inter-observer concordance in PD-L1 scoring. Digital image analysis software and ongoing quality assurance programs are critical for ensuring consistent and accurate interpretation of PD-L1 status. Online training tools [53] and digital pathology platforms [76, 77] have been developed to assist pathologists in PD-L1 interpretation and improve the accuracy and reproducibility of PD-L1 scoring. Automated scoring methods may also accelerate PD-L1 testing and alleviate the burden on pathologists [76, 78]. The increasing use of digital pathology and AI in routine practice provides an opportunity to leverage these tools to improve the clinical value of PD-L1 expression in NSCLC.

6 | Clinical Value of AI-Assisted PD-L1 Scoring Methods

6.1 | AI Models Optimize PD-L1 Clinical Assay Implementation

AI models can perform tasks that require human intelligence, including visual perception, decision making, communication, learning, and problem solving [79]. The improvement of

automated PD-L1 scoring using AI-based models is overcoming some of the limitations of manual scoring by pathologists using standard PD-L1 scoring methods and increasing the efficiency of pathology workflows and scoring throughput. Several studies have demonstrated the feasibility of using AI models to automate digital image analysis and help pathologists analyze PD-L1 expression and predict response to ICIs. We summarize the performance of these models in Table 4. Overall, the results of these studies are encouraging, demonstrating that automated PD-L1 expression analysis using AI models provides high specificity and accuracy in determining PD-L1 expression on tumor cells comparable to manual scoring by trained pathologists and can improve the efficiency of PD-L1 scoring [80–87].

6.1.1 | AI Model Scoring Exploration

Cheng et al. [82] developed a deep learning model to analyze PD-L1 expression in IHC-stained tissues from 1288 patients with lung cancer. Tissues were stained for PD-L1 using both the 22C3 and SP263 assays. The AI model showed high performance in scoring PD-L1 expression in tissues stained with 22C3, especially at TPS $\geq 1\%$. High accuracy and specificity were also achieved when the AI model was used to determine PD-L1 expression in tissues stained with SP263. The PD-L1 scoring results with AI were consistent with those obtained manually by a pathologist [82]. These results suggest that AI models could help automate PD-L1 scoring in IHC-stained NSCLC tissues.

Aitrox (a new AI segmentation model) [83] and AIM-PD-L1-NSCLC (a scanner- and antibody-agnostic machine learning model) [84] have also been tested as AI-assisted PD-L1 scoring systems for NSCLC. Aitrox was trained on 54 whole slide images (WSIs) [83], and AIM-PD-L1-NSCLC was trained to identify PD-L1-positive cells in digitized WSIs of NSCLC tissue samples collected at various institutions [84]. When used to score PD-L1 expression in IHC-stained NSCLC images, Aitrox provided PD-L1 scores similar to those obtained by experienced pathologists [83]. In addition, Aitrox performed better than inexperienced pathologists in determining PD-L1 scores [83]. AIM-PD-L1-NSCLC provided PD-L1 scores that were highly concordant with those obtained by board-certified pathologists, regardless of the PD-L1 antibody clone used to stain the tissues [84].

6.1.2 | AI Models for Efficacy Prediction and Biomarker Screening

AI models have also been tested for their ability to predict response to anti-PD-1/PD-L1 therapies. Lunit SCOPE PD-L1 is an AI model trained on 393,565 tumor cells manually annotated by a pathologist for PD-L1 expression in 802 WSIs stained using the 22C3 assay [88]. The model accurately predicted PD-L1 expression, with an AUC value of 0.889. High concordance rates in TPS between the AI model and the pathologist were observed for high TPS values (85.7% for PD-L1 $\geq 50\%$ and 89.3% for PD-L1 1%–49%). When stratified into two subgroups based on AI-predicted TPS ($\geq 1\%$ vs. $<1\%$), there was a significant

difference in median progression-free survival between the two patient groups (2.8 vs. 1.7 months; hazard ratio [HR]: 0.52; 95% CI: 0.38–0.71; $p < 0.001$) [88]. Although both manual and AI-powered prediction of PD-L1 scores can predict response to ICIs, studies suggest that patient stratification based on AI-derived PD-L1 scores can result in improved associations with survival after treatment with ICIs [79, 88].

AI has also been used to predict response to ICIs by combining PD-L1 with other biomarkers. Althammer et al. [78] used AI to score digital images for PD-L1+ and CD8+ cell densities and developed a CD8xPD-L1 signature to mark NSCLC samples with high PD-L1+/CD8+ cell densities. They found that median overall survival after treatment with durvalumab was longer for CD8xPD-L1 signature-positive patients than for signature-negative patients (21.0 vs. 7.8 months; $p = 0.00002$) [78]. These findings suggest that the use of AI to identify NSCLC samples with high PD-L1+/CD8+ cell densities could aid pathologists in selecting patients who are likely to respond to durvalumab therapy.

6.2 | Limitations and Challenges of AI-Assisted PD-L1 Scoring Methods

Although AI has shown promise in increasing the accuracy of PD-L1 scoring in NSCLC, the technology is relatively new, and several technological challenges must be addressed (Figure 2). Firstly, almost all published studies evaluating the feasibility and performance of AI-assisted PD-L1 scoring methods are retrospective (Table 4). Large prospective studies are needed to validate the ability of AI models to accurately quantify PD-L1 expression in digitized NSCLC tissue images and predict response to anti-PD-1/PD-L1 therapies. Secondly, although AI models can improve objectivity and reduce certain human-specific biases [91], they can also introduce other types of bias. For example, confounding effects, such as those introduced during the preparation of sample slides, as well as imbalances in the training datasets, can introduce biases in AI-assisted decisions [92]. Underfitting and overfitting are two common issues in training and evaluating AI models that have a significant impact on bias and variance, especially due to a lack of high-quality, extensively labeled pathology images. Underfitting occurs when a model is too simple to capture the underlying patterns in the data, leading to high training and testing errors due to high bias and low variance [93]. On the other hand, overfitting occurs when a model is excessively complex, capturing noise in the training data and resulting in low training error but high testing error due to low bias and high variance. Most published studies evaluating the use of AI to predict PD-L1 expression or response to immunotherapy are small, and AI training is particularly prone to overfitting when using small datasets. Image segmentation algorithms using AI may be helpful in reducing human-specific biases and other biases (Huang et al. in preparation). Ethical and legal concerns of AI should also be taken into consideration. Many uncertainties remain regarding the integration and management of patient data when AI systems are used to reach a diagnosis or make treatment decisions [94, 95]. Key ethical issues to consider when using AI to score PD-L1 expression or identify patients who may benefit from treatment with ICIs

TABLE 4 | Performance of AI models in predicting PD-L1 expression and treatment response in non-small cell lung cancer.

Study	Design	Samples	Task	Metric, cutoff	AUC	ICC (95% CI)	Accuracy (%)	Sensitivity (%)	Specificity (%)	Association with ICI response
Naso et al. 2021 [80]	Retrospective	IHC-stained tissues	PD-L1 scoring	TPS, ≥ 1%	NR	0.80 (0.71–0.88)	NR	95	84	NR
Pan et al. 2021 [81]	Retrospective	IHC-stained tissues	PD-L1 scoring	TPS, ≥ 1%	NR	SP263: 0.873 (0.835–0.903) 22C3: 0.737 (0.641–0.810)	NR	NR	NR	NR
Cheng et al. 2022 [82]	Retrospective	IHC-stained tissues	PD-L1 scoring	TPS, ≥ 1%	NR	NR	96.4	94.07	95.02	NR
Huang et al. 2022 [83]	Retrospective	IHC-stained tissues	PD-L1 scoring	TPS	NR	NR	TPS < 1%: 85.29 TPS 1%–49%: 77.97 TPS > 50: 72.73	NR	NR	NR
Griffin et al. 2022 [84]	Retrospective	IHC-stained tissues	PD-L1 scoring	TPS	NR	0.93 (0.90–0.94)		NR	NR	NR
Wu et al. 2022 [85]	Retrospective	IHC-stained tissues	PD-L1 scoring	TPS	NR	0.866 (0.787–0.922)	0.9326	NR	0.9641	NR
Lim et al. 2022 [86]	Retrospective	312 PET/CT scans (189 adenocarcinomas, 123 squamous cell carcinomas)	Prediction of PD-L1 expression	TPS, ≥ 50%	0.712 ^a	NR	NR	75.3	58.2	NR
Zhao et al. 2023 [87]	Retrospective	334 PET/CT scans	Prediction of PD-L1 expression	TPS, ≥ 1%	0.761 ^a	NR	70.30	68.52	72.34	NR
Kim et al. 2021 [88]	Retrospective	IHC-stained tissues	Prediction of PD-L1 expression and response to ICIs	TPS, ≥ 1%	0.889 ^b	NR	NR	NR	NR	mPFS in TPS ≥ 1% versus < 1%: 2.8 versus 1.7 months (HR: 0.52 [95% CI]: 0.38–0.71; <i>p</i> > 0.001)

(Continues)

TABLE 4 | (Continued)

Study	Design	Samples	Task	Metric, cutoff	AUC	ICC (95% CI)	Accuracy (%)	Sensitivity (%)	Specificity (%)	Association with ICI response
Baxi et al. 2022 [79]	Retrospective	IHC-stained tissues	Prediction of PD-L1 expression and response to ICIs	TC, ≥ 1% or ≥ 5%	0.602 ^b	NR	NR	NR	NR	ORR in TC < 1% versus ≥ 1%: 5.3 versus 21.1 (OR: 5.08 [95% CI: 0.65–39.55]) ORR in TC < 5% versus ≥ 5%: 9.5 versus 25.8 (OR: 3.53 [95% CI: 0.79–15.67])
Althammer et al. [78]	Prospective	IHC-stained tissues	Scoring for PD-L1+/CD8+ cell densities, prediction of response to ICIs	NR	NR	NR	NR	NR	NR	mOS in CD8xPD-L1 signature-positive versus signature-negative: 21.0 versus 7.8 months (<i>p</i> = 0.00002)
Vaidya et al. 2020 [89]	Retrospective	Pretreatment CT scans	Predict risk of hyperprogression	NR	0.96 ^b	NR	NR	NR	NR	HR for OS: 2.66 [95% CI: 1.27–5.55]; <i>p</i> = 0.009
Mu et al. 2021 [90]	Retrospective	697 PET/CT scans	Prediction of PD-L1 expression and response to ICIs	TPS, ≥ 1%	0.84 ^b	NR	78.45	77.43	81.48	HR for PFS in DLS ≥ 0.55 versus < 0.55: 0.41 (95% CI: 0.25–0.67), <i>p</i> = 0.001 HR for OS in DLS ≥ 0.55 versus < 0.55: 0.48 (95% CI: 0.25–0.91), <i>p</i> = 0.024

Abbreviations: AI, artificial intelligence; AUC, area under the receiver operating characteristic curves; CI, confidence interval; DLS, deeply learned score; HR, hazard ratio; ICC, intraclass correlation coefficient; ICI, immune checkpoint inhibitor; IHC, immunohistochemistry; mOS, median overall survival; mPFS, median progression-free survival; NR, not reported; OR, odds ratio; ORR, objective response rate; PD-L1, programmed death-ligand 1; PET/CT, positron emission tomography/computed tomography; TC, tumor cell; TPS, tumor proportion score.
^aFor predicting PD-L1 expression.
^bFor predicting objective response.

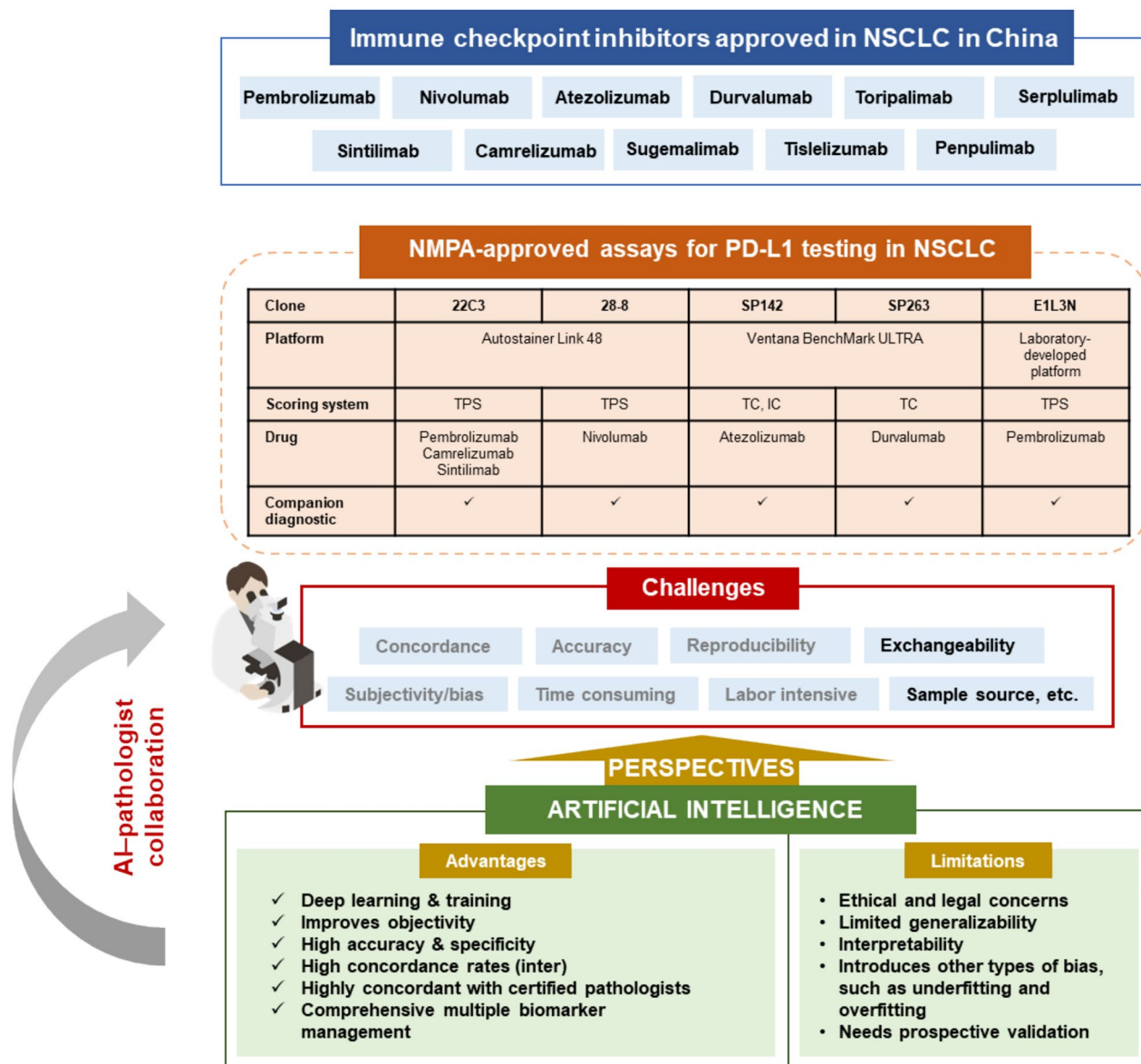


FIGURE 2 | Summary of advantages and challenges for using artificial intelligence models to assist PD-L1 scoring illustration: DataBase Center for Life Science (DBCLS), CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>), via Wikimedia Commons. IC, immune cell; NMPA, China National Medical Products Administration; NSCLC, non-small cell lung cancer; PD-L1, programmed death-ligand 1; TC, tumor cell; TPS, tumor proportion score.

include privacy, choice, equity, trust, and assigning responsibility if errors are made [95]. Currently, risk regulation is increasingly used to govern AI systems, focusing on managing the risks associated with AI rather than outright banning its use. Determining who has responsibility when AI makes a mistake involves a multifaceted approach including updating legal frameworks, implementing hybrid liability regimes, and ensuring transparency and accountability. Ongoing efforts by the EU and other international bodies are crucial steps toward addressing these challenges and ensuring that AI technologies are used responsibly and ethically [96, 97]. Laws, regulations, and robust public governance mechanisms and ethical frameworks are required to ensure the ethical use of AI models and protect patient privacy. These are reasons for insufficient high-quality massively labeled training samples for AI pathology models.

The limited generalizability across patient populations due to small sample sizes, data imbalances, and heterogeneity between datasets are further barriers to the clinical implementation of AI-assisted diagnostic methods [79, 94]. In most published studies, AI models were trained using relatively small datasets; thus, the performance of AI-assisted PD-L1 scoring methods in underrepresented populations or in patients with rare tumor subtypes remains unknown [94]. Extensive training on large datasets comprising images from diverse patient populations and samples representing different histological and molecular subtypes and clinical stages is needed to increase the generalizability of AI-assisted PD-L1 scoring models. In addition, AI models have shown limited robustness in certain settings. Validation studies have confirmed that AI models perform as well as or better than pathologists in most settings [82, 85]; however, their performance

is poor when used to analyze images with low PD-L1 expression (52.4% for images with TPS < 1%) [88], especially those with large regions of false-positive cells [83]. Therefore, AI algorithms should be optimized, extensively validated, and reviewed before their implementation, and their performance should be verified to ensure that they are safe, efficacious, and reliable.

Interpretability and access to technology should also be improved to fully realize the potential of AI to support treatment decisions for patients with NSCLC. The limited interpretability of AI, also referred to as the “black box”, is a critical issue that hinders the widespread clinical adoption of AI-assisted systems [94]. Training programs on AI are warranted to ensure that pathologists know how to use AI models and interpret AI-assisted PD-L1 testing results.

Access to digital pathology in China is currently limited and faces several challenges. Digital pathology was initiated relatively late in China and has progressed more slowly compared to other countries. There is currently no comprehensive implementation of complete digital pathology (CDP) in China [98]. Despite this, the development of digital pathology has been accelerated through advances in technologies such as the internet, big data, and AI. In addition, the Chinese healthcare industry is leveraging cloud and mobile computing to enable better collaboration and access to medical information. Government support and assistance from international companies are also playing a role in the development of digital pathology. Indeed, in the present day, China is a key international contributor to the development of digital pathology, particularly in lung cancer. For example, the integration of whole-slide imaging technology with AI algorithms is particularly significant in the context of lung cancer, where it enhances the accuracy of classification and prediction. However, most research in China is conducted independently, highlighting the need for stronger academic collaboration and data sharing between nations [99]. There are also significant challenges related to cognition, classification, and quality that need to be addressed. The future development of digital pathology in China will likely involve the integration of AI and big data to enhance diagnostic capabilities and personalized medicine. This will require addressing current barriers and leveraging international experiences and technologies [98, 99].

Last but not least, AI models require extensive validation before clinical implementation in China and have their own challenges. For example, the cost and accessibility of AI technology may limit its widespread implementation in clinical practice [100]. AI in pathology in China is currently in the research stage [101], and investing in AI-related software and hardware can be expensive. Efficient and high-quality scanners are needed to ensure that the implementation of AI will improve work efficiency. However, the degree of automation of instruments currently available in most pathology departments in China is not high [56].

6.3 | AI-Pathologist Collaboration and Enhancement

Although AI has shown promise in increasing the accuracy and reproducibility of PD-L1 scoring in NSCLC, AI models are

unlikely to replace pathologists in PD-L1 scoring [102, 103]. Instead, pathologists will be aided by AI models to accurately and quickly analyze large datasets of digitized tissue images to determine PD-L1 expression and make treatment decisions. In this workflow, pathologists play a key role in the operation and interpretation of the technology. However, some aspects of their work may change because tedious, time-consuming, and manual tasks can be automated using AI.

In this collaborative model, pathologists play a crucial role in training the AI algorithm using manually annotated training images, as well as in performing quality control by assessing the accuracy and reproducibility of results [102, 103]. Pathologists also play a critical role in the validation, implementation, and interpretation of AI-assisted systems, ensuring their accuracy and reliability in clinical practice [104]. AI models can also be used to confirm PD-L1 testing results obtained through manual scoring by a pathologist. The College of American Pathologists has developed guidelines for the validation of WSI systems for diagnostic use [105], according to which pathologists play a crucial role in validating and verifying the performance of these systems to ensure their suitability for clinical use [105].

Training and education are essential to enable pathologists to use digital pathology systems effectively to increase their productivity, reduce diagnostic turnaround times, increase the clinical value of PD-L1, and reduce diagnostic errors. The Royal College of Pathologists has provided guidance for digital pathology implementation, which includes recommendations for training and validation [106]. Pathologists need to be trained in the use of digital pathology systems, including image acquisition, interpretation, and analysis [106]. Pathologists also play a role in training and mentoring other healthcare professionals in the use of digital pathology technology [106]. Although a nationwide telepathology consultation and quality control program was implemented in China in 2014 [107], Chinese guidelines on the implementation of AI-assisted PD-L1 scoring models and other digital pathology systems are currently lacking.

In summary, AI, along with automation, has the potential to improve workflow efficiency and address pathologist capacity limitations by aiding in diagnostic tasks and automating routine processes. For instance, AI-based image analysis can help mitigate staffing shortages by improving diagnostic efficiency and standardizing evaluations [108, 109]. However, education and experience with AI are needed to support its implementation across allied health professions [110]. We believe that training and partnerships with academic institutions can help prepare the workforce for AI integration.

7 | Prospects

Accurate evaluation of PD-L1 expression is critical for identifying patients with NSCLC who are most likely to benefit from immunotherapy. Conventional PD-L1 testing is limited by subjectivity, variability, and a lack of standardization. Recent advances in AI have led to the emergence of numerous AI-assisted PD-L1 scoring methods, which could help overcome some of the limitations of standard PD-L1 testing methods. AI-based models have shown promise in automating PD-L1 scoring while

providing high specificity and accuracy. In addition, although PD-L1 expression is an important biomarker for immunotherapy response, ongoing exploration and development of additional predictive biomarkers are necessary to obtain a more comprehensive understanding of tumor-immune interactions and improve patient selection for immunotherapy. PD-L1 expression can be dynamic and heterogeneous within tumors, leading to sampling bias and potential misclassification [75]. Various studies have explored predictive biomarkers beyond PD-1/PD-L1 receptors to enhance the prediction of immunotherapy response in NSCLC [111–114]. Integrating additional biomarkers, such as TMB, gene mutations, microsatellite instability, HLA expression, and IFN γ level with PD-L1 expression may enhance the prediction of immunotherapy response.

It is important to note that the currently available AI tools for pathology have flaws and limitations. Ongoing research on improving current AI models, harmonizing testing methodology, minimizing technical and interpretation variability, correlating results with clinical outcomes, and addressing ethical concerns is essential to optimize the use of ICIs in the treatment of NSCLC in China. Further research is necessary to refine and validate AI-powered PD-L1 scoring methods to ensure their safe, effective, and reliable use in clinical practice. Generative AI tools are also gaining popularity, and future studies are needed to explore their potential use in supporting clinical decisions for patients with NSCLC. Only through such efforts can the promise of precision immunotherapy based on PD-L1 status be fully realized for Chinese patients with advanced NSCLC.

Author Contributions

Conceptualization: Ziling Huang, Shen Wang, Jiansong Zhou, Haiquan Chen, and Yuan Li. **Writing – Original Draft Preparation:** Ziling Huang, Shen Wang, and Jiansong Zhou. **Writing – Review and Editing:** Ziling Huang, Shen Wang, Jiansong Zhou, Haiquan Chen, and Yuan Li. **Supervision:** Haiquan Chen and Yuan Li. All authors read and approved the final version of the manuscript.

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Conflicts of Interest

Jiansong Zhou reported that he was an MSD China employee at the time of writing. Other authors have no conflicts of interest to declare.

Data Availability Statement

All data reported in this review were sourced from the cited references.

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